

Model Building For U.S. Census Bureau

Data Exploration & Pre-Processing

The dataset contains approximately 200K census survey records with 40 demographic and employment-related variables and a binary income label (>50K vs <=50K). The target distribution is highly imbalanced, with roughly 94% belonging to the low-income class. Initial exploration identified a mix of:

1. Continuous numeric attributes (age, capital gains, wage per hour etc.)
2. Categorical attributes (sex, race, marital stat etc.)
3. Integer-coded categorical attributes (detailed occupation recode, veteran benefits etc.)
4. Some highly skewed attributes (capital gains, capital losses etc.)
5. Migration related attributes which are dominated by "Not in universe"

Some Observations:

1. Education level strongly correlated with income
2. Executive/professional occupations have higher income concentration
3. Self-employed incorporated workers show elevated high-income rates
4. Strong gender imbalance exists in high-income outcomes

Preprocessing Approaches

For classification:

1. The label column was converted into binary format (1 = >50K, 0 = <=50K)
2. The migration-related variables are dropped due to high missingness and limited business interpretability
3. The categorical variables are one-hot encoded
4. The numeric variables are median-imputed
5. The logistic regression used feature scaling
6. The class_weight='balanced' is used for class imbalance problem and later results and impact are shown in various plots for precision-recall effect.

For segmentation:

1. The income label is removed from clustering inputs
2. The survey weight is also removed from the clustering algorithm.
3. The highly skewed monetary features are log-transformed
4. The rare categorical levels are collapsed into "Other"
5. The numeric variables are standardized
6. The categorical variables are one-hot encoded

Model Training & Evaluation

Three classification approaches were evaluated:

1. Logistic Regression
2. Random Forest
3. XGBoost

Logistic Regression is considered a baseline model.

Random Forest is used to capture nonlinear relationships and feature interactions. Both balanced and imbalanced versions are evaluated to analyze precision-recall tradeoffs.

XGBoost is selected as the strongest tabular learner due to:

- superior ROC-AUC
- better minority-class F1-score
- improved precision-recall balance

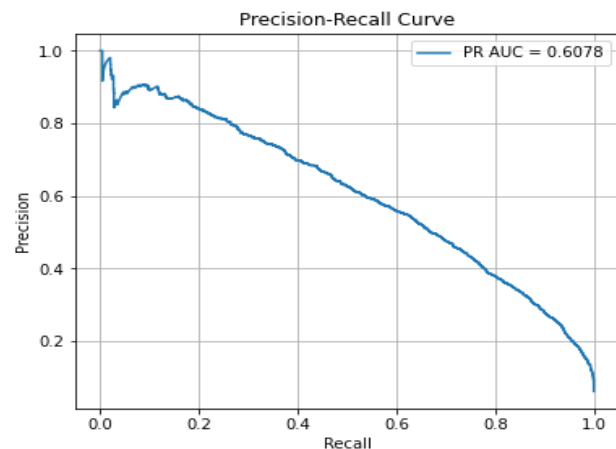
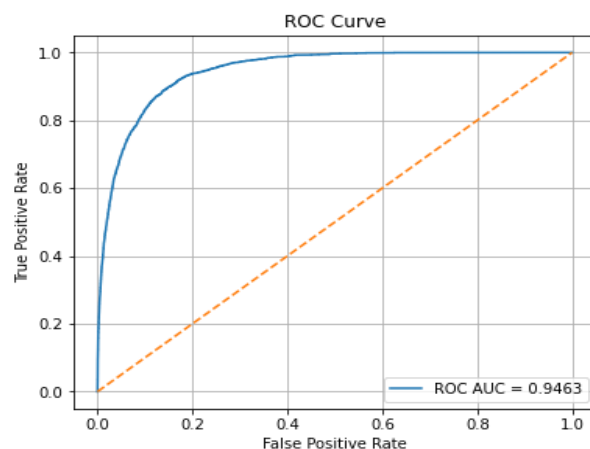
Training used:

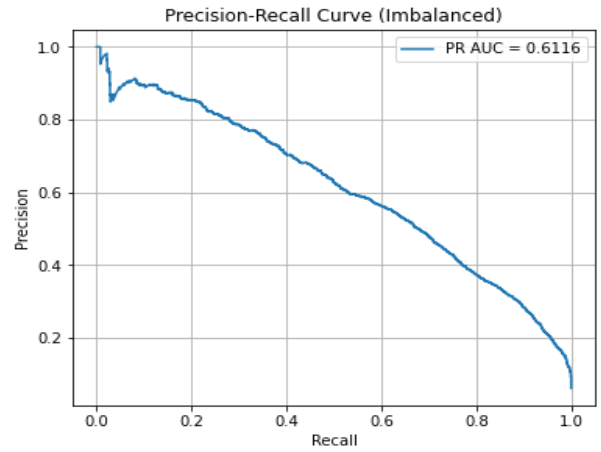
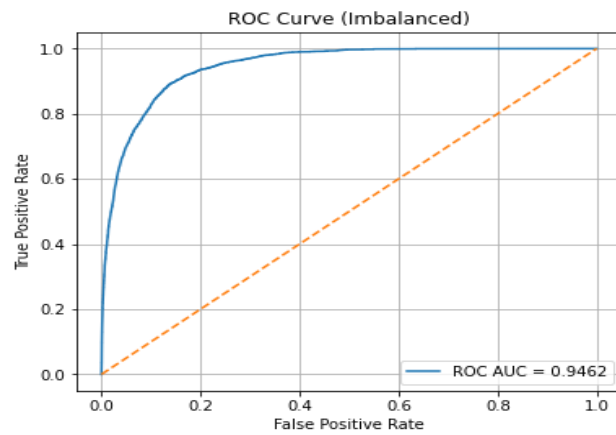
- stratified train-test split
- scikit-learn preprocessing pipelines
- probability-based evaluation metrics

Classification Model Results

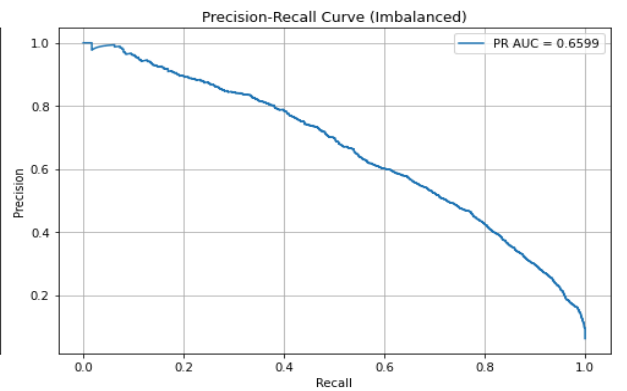
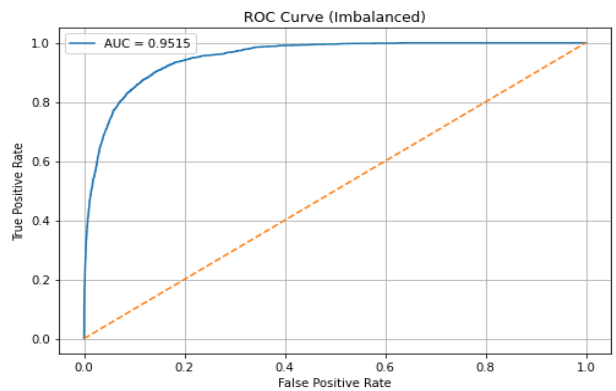
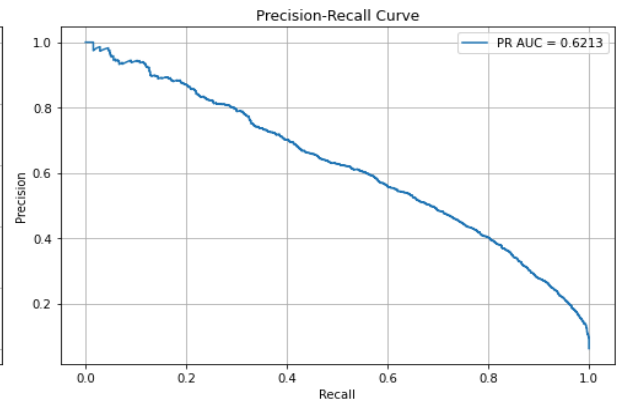
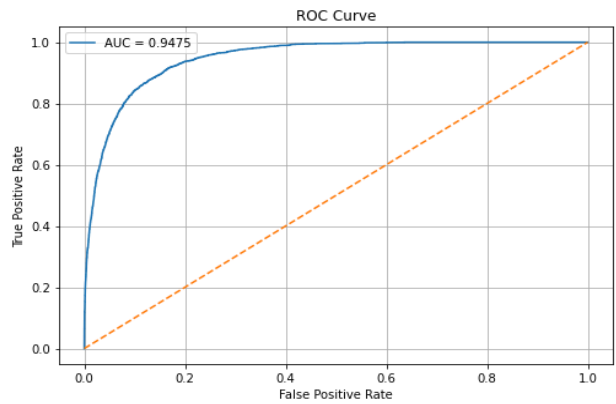
Comparative Plots & Table For Classification Model:

1. Logistic Regression

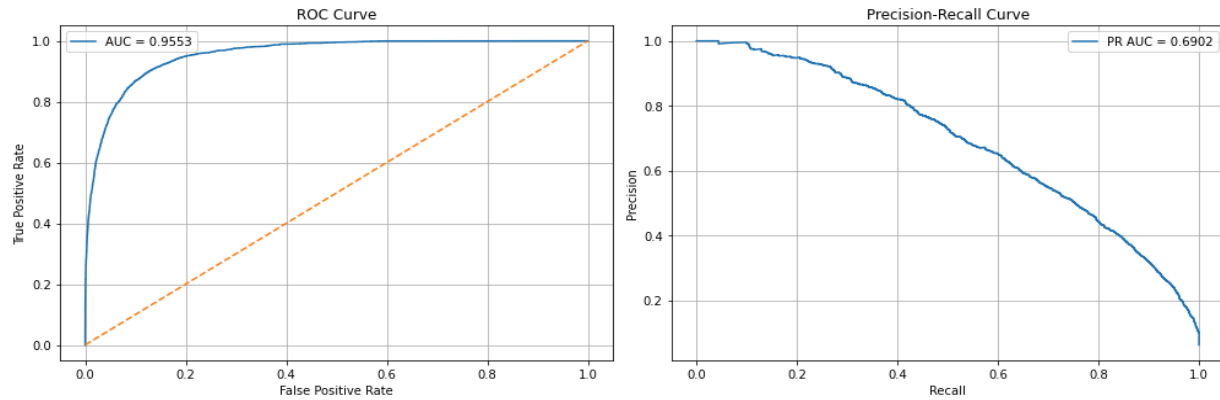




2. Random Forest



3. XGBoost



Model Class 1 = "50000+." Class 0 = "- 50000."	Acc	ROC-AUC	Precisi on (Class 0)	Precisi on (Class 1)	Recall (Class 0)	Recall (Class 1)	F1 (Class 0)	F1 (Class 1)
Logistic Regression (Imbalanced)	0.95	0.9462	0.96	0.73	0.99	0.38	0.98	0.50
Logistic Regression (Balanced)	0.85	0.9463	0.99	0.29	0.85	0.90	0.92	0.44
Random Forest (Imbalanced)	0.96	0.9515	0.96	0.81	0.99	0.36	0.98	0.50
Random Forest (Balanced)	0.93	0.9475	0.98	0.46	0.94	0.74	0.96	0.56
XGBoost	0.96	0.9553	0.97	0.77	0.99	0.46	0.98	0.57

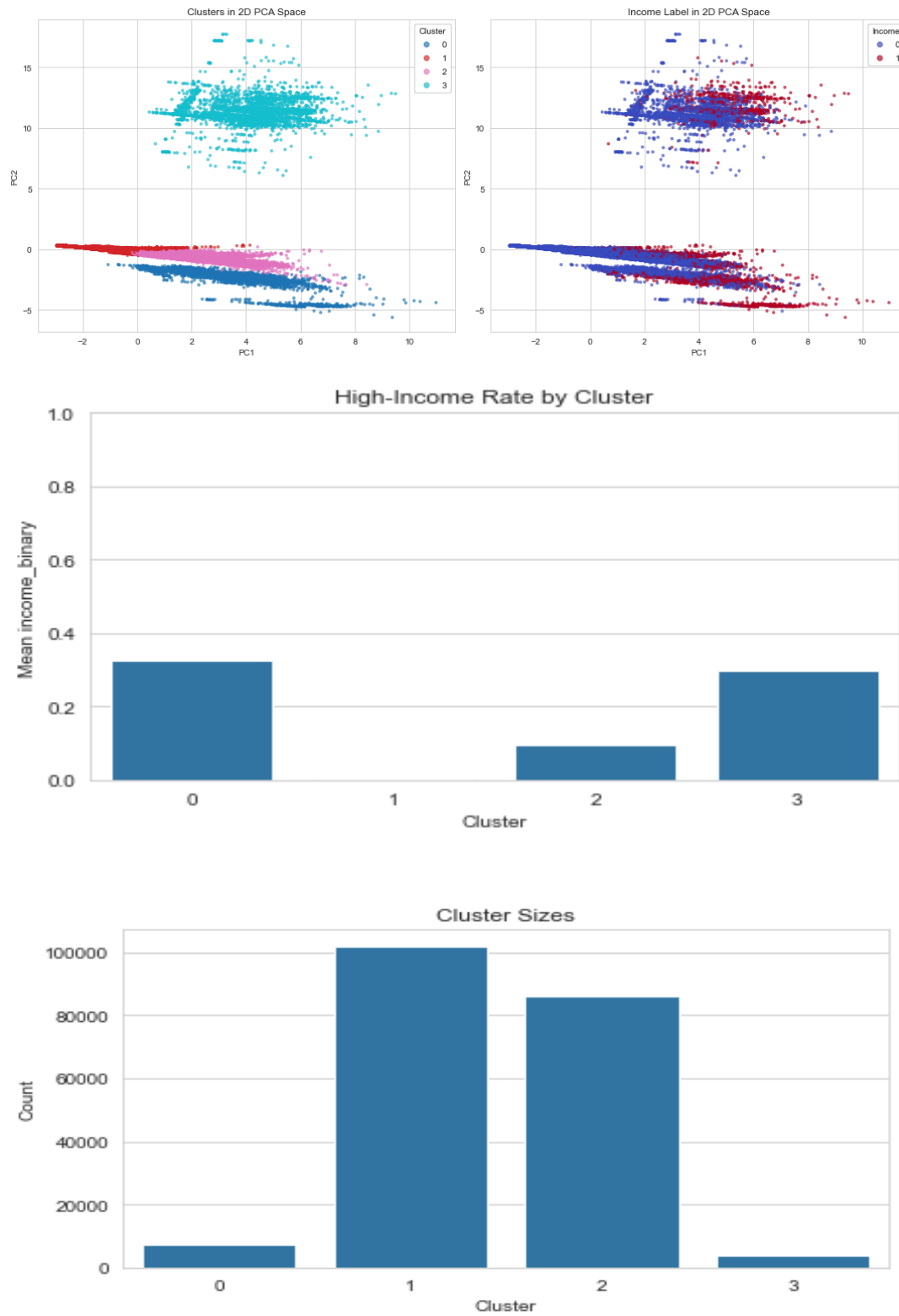
Classification Model Observations:

1. Since the dataset is highly imbalanced, accuracy alone could not be a good measure of the model. Hence, I explored ROC-AUC and precision-recall curves for clarity.
2. Based on the experiment results so far using Logistic regression, random forest, and XGBoost, I would investigate XGBoost for threshold tuning to better understand the trade-off.
3. The Precision-Recall Trade-off can be summarized as follows:

Classification Model	Pros	Cons
RF Imbalanced	High precise	Misses many high-income individuals
RF Balanced	High recall	Too many false positives
XGBoost	Best balance overall	Still need improvement

4. Logistic Regression could be a decent interpretable baseline.
5. Random Forest could be a decent nonlinear ensemble baseline.
6. XGBoost could be a decent tabular learner with best overall balance for further tuning based on additional constraints. It has highest ROC-AUC and better performance on structured tabular data, fewer false positives than balanced RF and better recall than imbalanced RF.

Segmentation Model Results



Segmentation Observation:

KMeans clustering identified four meaningful population segments:

- The affluent professional groups
- The middle-income working households
- The economically inactive/dependent populations

Result Interpretation:

1. Cluster 1 is a lower income segment. Probably dominated by “Not in universe” having the majority of children or never-married individuals. This could be the population of dependents, students, and economically inactive people. This cluster would be suitable for low-cost awareness, and family oriented offers.
 - a. Outcome: Low cost family-focused campaigns
2. Cluster 2 is a working segment having the highest share of private-sector workers, a large share of high-school graduates and mixed occupation profile with administrative, professional and service roles. The income rate is not high but not zero either which makes it suitable for value-focused segments such as everyday products, financing options, and discounts.
 - a. Outcome: Middle class or working class offers and marketing strategies.
3. Cluster 0 and 3 are the premium marketing target segments having similar gender distribution along with higher concentration of education, professional degrees and high-income rate. Cluster 0 seems to be married-civilian-spouse, private-sector worker with heavy concentration of professional jobs. Cluster 3 is similar but more executive and slightly self-employed with bachelor and high school degrees. These segments should be prioritized for premium cards, investment products, high-end retail, travel, and loyalty programs.
 - a. Outcome: Premium and high-value marketing strategies.

Detailed tables for above mentioned points are in the notebook.

Marketing Recommendations:

- The affluent clusters should receive premium product targeting and loyalty campaigns.
- The middle-income clusters are suitable for value-oriented marketing.
- The economically inactive clusters are less suitable for premium targeting and better aligned with low-cost promotional strategies.